

Newman Conformations

1. Draw the staggered conformation and eclipsed conformation Newman projections for ethane.
2. (Note the dihedral angle). Draw the 0° Eclipsed, 60° Gauche, 120° Eclipsed, 180° Anti, 240° Eclipsed, and 300° Gauche conformations for butane.

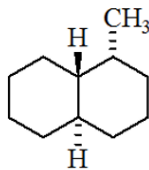
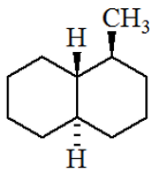
Chair Conformations

1. Draw a chair conformation for each molecule below. Draw the ring-flipped version of each molecule. Circle the more stable of the two conformers.

cis-1-Bromo-2-chlorocyclohexane
trans-1,3-diethylcyclohexane
2. Draw the more stable chair conformation for each of the following cyclohexanes. Then "flip" the ring and redraw the molecule in the higher energy form.

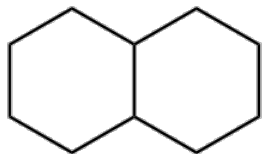
chlorocyclohexane
trans-1-methyl-3-propylcyclohexane
cis-1-chloro-2-methylcyclohexane
cis-1-tert-butyl-4-methylcyclohexane

3. Which compound below is more stable? Explain.



4.

There are two possible arrangements for decalin.



decalin

- Draw out the two stereoisomers.
- Which is the most stable and why?

5.

Glucose is a simple sugar with 5-substituents bonded to a 6-membered ring.

- Using a chair representation, draw the most stable arrangement of these substituents on the 6-membered ring.
- Convert the representation into one that uses a hexagon with wedges and dashes (2D structure).

